Support co-pilot: worth running?

VALETAX CASE | 10,000 tickets/month | Planning scenario | 14 October 2026

Recommendation: run a bounded, human-reviewed pilot.
Start with Sonnet. Select a cheaper model or cascade only after the same safety and quality evaluation, then measure operator time. No application or live evaluation has been run yet.

1. Compare configurations at equal quality

Assume 2,500 input + 500 output tokens per call, standard global API, no caching/batch discount. Sonnet: $2/$10 per million input/output tokens; Haiku: $0.10/$0.50 for short prompts. Include a 10% usage reserve. A cascade pays for both calls on fallback tickets. Actual usage must replace these estimates.

Configuration	LLM/month	Full OPEX/month	Net value/month
Sonnet 5.5	$110.00	$530.00	$5,670.00
Haiku 5.5	$5.50	$425.50	$5,774.50
Haiku + 20% Sonnet	$27.50	$447.50	$5,752.50

Full OPEX includes estimated infrastructure $100/month and maintenance 8h x $40 = $320/month. One-time integration allowance: 40h x $40 = $1,600, separate from the 3-5h take-home. Add any incremental licenses, compliance work and on-call costs. Prices are pre-tax.

2. Value comes from net operator time saved

Assumptions: manual handling 8 min; draft attempted on 60% of tickets; 80% of those drafts accepted; accepted drafts save 4 min **including review/editing**; rejected drafts add 0.5 min; fully loaded operator cost $20/h. Uncovered tickets receive no time-saving credit. All tickets are charged in the LLM budget.

Net hours = 10,000 x 0.60 x [0.80 x 4 - 0.20 x 0.5] / 60 = 310h/month.
Capacity value = 310 x $20 = **$6,200/month**; Sonnet net value = $6,200 - $530 = **$5,670/month**. This is capacity value, not automatic payroll savings; cash benefit requires avoided hiring or another operational change.

3. Sensitivity changes the decision

Conservative case (30% attempted, 60% accepted, 1 min saved, 0.5 min wasted) yields only $400/month: **Sonnet net = -$130/month**. With base assumptions, Sonnet breaks even above 27.4 seconds saved per accepted draft, or about 690 tickets/month at unchanged fixed costs. Haiku saves only $104.50/month versus Sonnet; an average loss of 1.88 seconds per incoming ticket erases that saving. Model quality is unmeasured, so the table is not a model ranking.

4. Pilot gates and scale path

Measure assisted vs manual active handling time on comparable tickets, draft acceptance, repeat contacts, citation support, specialist-routing recall, actual tokens and p95 latency. Go only with positive measured value and no observed critical violations in the agreed evaluation. Never give trading advice or decide complaints/KYC/AML; keep human approval. Scale with ticketing integration, durable queues and versioned KB; introduce retrieval when corpus growth warrants it.

Source: Anthropic API pricing, checked 14 October 2026. Volume and hard rules: supplied assignment. All other operational/economic values are assumptions.